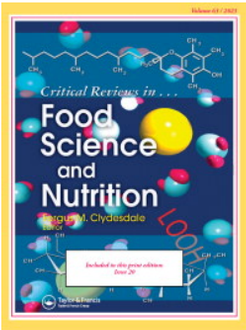

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REVIEW ARTICLE

Food-derived bioactive peptides as momentous food components: Can functional peptides passed through the PI3K/Akt/mTOR pathway and NF-κB pathway to repair and protect the skeletal muscle injury?

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ABSTRACT

Muscle injury is defined as an overuse injury or traumatic distraction of a muscle, which is latent in any sport event, from amateur to large events. Based on previous numbers of muscle injuries and time spent to the athletes' recovery, the use of dietary functional factors intervention strategies is essential to enhance the recovery process and health. In recent years, there has been increasing evidence that biologically active peptides played an important role in sports nutrition and muscle injure recovery. Food-derived bioactive peptides were physiologically active peptides mostly derived from proteins following hydrolysis, which could be resorbed in intact form to reduce muscle damage following exercise and induce beneficial adaptions within the connective tissue. However, the complexity of the histoarchitectural considerations for skeletal muscle injuries and the repair mechanism of damaged skeletal muscle were not well known. In the following overview, the potential mechanisms and possible limitations regarding the damaged skeletal muscle metabolism were summarized, which aimed to present an overview of the nutritional strategies and recommendations after a muscular sports injury, emphasizing the use of main bioactive peptides. In addition, this review will provide implications for the studies of dietary bioactive peptides in the future.

KEYWORDS

Bioactive peptides; muscle injury; repair mechanism; oxidative stress; inflammation; cartilage

Introduction

Food-derived bioactive peptides, as the functional active fragments of proteins, are derived from plant and animal origins including potato, nuts, cereals germ, soybean, egg, dairy products and meat proteins following chemical, enzymatic and microbial hydrolysis, which generally included 2–20 amino acids residues per molecule and the molecular masses are less than 6000 Da (Konig et al. 2021). After hydrolysis, food-derived bioactive peptides could be absorbed by intestine and enter the blood stream to interact with endogenous proteins or target receptor such as glycogen synthase or glucose transporter type 4 and mammalian target of rapamycin (Karami and Akbari-Adergani 2019). In view of the above-mentioned reasons, food-derived bioactive peptides have outstanding contributions in metabolic regulation and modulation, which also are regarded as the most momentous health-promoting food ingredients to improve human health and prevent disease such as dyslipoproteinemia, hypertension, oxidative stress and inflammation (Kamdem and Tsopmo 2019). In recent years, there has been a lot more evidence demonstrated that food-derived bioactive peptides might also play a key role in skeletal muscle injuries. In the rat model with volumetric loss of masticatory muscle, SVVYGLR peptide had been applied to investigate the effects on the biological properties of skeletal muscle-specific myogenic cells (Tanaka et al. 2021). SVVYGLR peptide played a potent role in facilitating skeletal muscle regeneration by promoting the migration, and differentiation of myogenic precursor and progenitor cells. The research provided the usefulness of the clinical application of SVVYGLR peptide in skeletal muscle dysfunctions involving serious damage caused by injury or surgery and in congenital diseases causing deformed or poorly developed muscular tissue. Taking food-derived bioactive peptides has positive effects on skeletal muscle injuries aspects such as regeneration, performance and structural adaptations.

Skeletal muscle injuries are defined as the overuse or traumatic distraction injury of muscle, which can be classified in structural muscle injury (direct muscle injury) or functional muscle disorder (indirect muscle disorder). Structural muscle injury refers to the any acute direct muscle injury with macroscopic evidence of muscle fiber disorder and the functional muscle disorder is a painful muscle injury without evidence of muscle fiber injury (Quintero et al. 2018). Structural muscle injury may cause a host of biochemical changes in affected area, which can lead to

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increased production of reactive oxygen species and inflammatory cytokines, thus further degrading muscle protein and causing damage (Rebolledo, Acuna, and Brandan 2021). Owing to the increased incidence of osteoarthritis and long-term joint injuries from sport participation, there are new emerging interests in therapeutic interventions to improve structural muscle injury. Collagen peptides are produced by hydrolysis of porcine, marine, bovine and poultry collagen, which are synthesized by a high concentration of proline, glycine and hydroxyproline. For skeletal muscle, collagen is the crucial structural component of extracellular connective tissue, which can improve the synthesis rates of muscle protein as well as provide elasticity for tissues and transmit the contraction force to tendons and bones (Holwerda and van Loon 2022; Supruniuk, Górski, and Chabowski 2023). Khatri et al. (2021) mainly focused on the impact of collagen peptides on bone and joint health and their roles in bone turnover and regulating tendon. According to the hypothesis of "mechano-transduction", the mechanical loading of tendon tissue during exercise generates multiple signal cascades in tissue cells, thus increasing the production of proteoglycan, type *I*, *II*, *IV* collagen and elastin synthesis to reduce tissue damage and decreasing pain. The collagen synthesis may also increase with the co-ingestion of vitamin C, due to the role of vitamin C in the hydroxylation of lysine and proline, both of which are prominent to the formation of collagen helix as well as intermolecular cross-linking. Moreover, glycine can enhance collagen organization strength and inhibit the release of pro-inflammatory cytokines (e.g., interleukin-6) to influence tenocyte metabolism, making collagen peptides possess anti-inflammatory. However, the research mainly focused on the collagen from animal cartilage and bone but the collagen from vegetarian/vegan sources. Therefore, it is necessary to review the role of the food-derived bioactive peptides (vegetarian/vegan sources and animal sources) in skeletal muscle injuries and investigate the evidence to date as well as potential mechanisms of histoarchitectural considerations for skeletal muscle injuries.

According to the current state of knowledge, there are several problems about food-derived bioactive peptides and the role in attenuating symptoms and signs of skeletal muscle injury occurring high resistance exercise: (*i*) the potential mechanism of food-derived bioactive peptides has not been systematically clarified; (*ii*) food-derived bioactive peptides cannot effectively eliminate the pain of muscle injury, only relieve certain signs and symptoms; (*iii*) the optimal pretreatment period of skeletal muscle injury is ascertain dimly and (*iv*) the recommended nutritional management and its combination with drug therapy during functional recovery phase are not yet clear. In the following overview, the potential mechanisms and possible limitations regarding the damaged skeletal muscle metabolism were summarized, which aimed to present an overview of the nutritional strategies and recommendations after a muscular sports injury, emphasizing the use of main bioactive peptides. In addition, this review will provide implications for the studies of dietary bioactive peptides in the future.

Anti-inflammatory and antioxidant function of polypeptides in the repair of skeletal muscle injuries

Skeletal muscle injuries are common among the elite athletes, individuals from all fitness backgrounds and sedentary person with no formal training experience, which have received considerable attention in recent years. Unaccustomed, strenuous exercise usually involves eccentric action, which causes muscle injuries in a fiber-specific manner. This initial skeletal muscle injuries will lead to a large number of biochemical changes in the affected area, such as increased production of reactive oxygen species and inflammatory cytokines, which will further degrade the muscle proteins and lead to delayed damage. Sufficient dietary-protein provision may greatly improve muscle mass and strength, which also can assist the recovery process (Islam et al. 2022). After the intestinal digestion and absorption, protein hydrolysate (i.e., bioactive peptides) can enhance the binding of amino acids in skeletal muscle proteins and improve the protein synthesis rate through the change of signal pathways (Sun and Udenigwe 2020). The mechanism of polypeptides in the repair of skeletal muscle injuries will be reviewed in detail in the following aspects.

Bioactive peptides relationships to protein synthesis and degradation in skeletal muscle

After high-intensity and vigorous exercise, it is commonly accepted that skeletal muscle protein degradation and synthesis will expand. Skeletal muscle protein degradation may occur through three main pathways including calpain-dependent, ubiquitin-proteasome and autophagy/lysosome. The efficiency of mRNA translation into peptides (i.e., translation efficiency) as well as the translation mechanical quantity (the number of ribosomes) of per unit tissue (i.e., translational capacity) are associated with the protein synthesis (Mirzoev 2020). Bioactive peptides are increasingly accepted as a momentous factor in regulating muscle protein turnover. Consumption of exogenous bioactive peptides can increase amino acids in body, which can provide induction of anabolic environment and then promote tissue remodeling (Bo et al. 2021). In previous trials, it has been proved that fermented soybean protein peptides were rich in arginine, aspartic acid, glutamic acid, leucine, lysine and proline, which could conspicuously improve the expression of AMP-activated protein kinase and PGC1-α genes as well as proteins in mice. Fermented soybean protein peptides intake also could optimize the mitochondrial function in skeletal muscle cells, strengthen the energy production of skeletal muscle, and delay the pain of muscle injury (Fang et al. 2022; Fan, Jia, and Du 2023). The mechanism of action might be that branched-chain amino acids from fermented soybean protein peptides inhibited the accumulation of lactic acid in bodies, extend the energy source of skeletal muscle, reduce the decomposition of muscle glycogen and rapidly increase the blood glucose ratio. The insulin-like growth factor 1 could activate phosphotidylinositol-3-kinase (P13K)/protein kinase B (AKT)/mTOR pathway, and then strengthen the protein synthesis through AKT-mediated phosphorylation of TSC2 (an endogenous inhibitor of TORC1).

The PI3K/Akt/mTOR pathway was considered to be the major pathway of involving in muscle protein synthesis and skeletal muscle hypertrophy (Figure 1). P13K, as a pivotal factor upstream of mTOR signal pathway, was involved in adjusting a range of physiological processes and played a key role in metabolism, differentiation, proliferation and survival of numerous cells. AKT, as the downstream signal molecule of PI3K, was a significant anti-apoptotic regulator and could mediate various biological effects. AKT-mediated phosphorylation of TSC2 could lead to the accumulation of Ras homologue rich in brain, which was also an effective TORC1 activator. Activated TORC1 effected the protein synthesis through phosphorylated ribosomal protein S6 kinase and eukaryotic translation initiation factor 4E-binding protein 1, which showed inhibition due to the mTORC1-dependent phosphorylation and removed the protein from its binding to eukaryotic translation initiation factor 4E allowing translation initiation to initiate (Danieli et al. 2023). The phosphorylation of phosphorylated ribosomal protein S6 kinase activated its kinase ability, causing it to phosphorylate ribosomal protein S6 (a component of the 40S ribosomal subunit), as well as activated protein translation (Hodson and Philp 2019). When activated, mTORC1 phosphorylates proteins involved in autophagy (unc-51 like autophagy activating kinase (ULK1), UV radiation resistance-associated gene protein and transcription factor EB (TFEB)) and the ubiquitin proteasome (extracellular signal-regulated kinase 5) systems, which inhibited the effects on autophagy formation and proteasome assembly, respectively, and inhibited the protein catabolism of the second component of skeletal muscle net protein balance. The inhibitory activity was determined by the following formula (1):

$$\%\text{Inhibition} = \frac{A_c\text{-}(A_s - A_b)}{A_c} \times 100 \quad (1)$$

where A_c was the control absorbance (absence of bioactive peptides), A_s was the absorbance of sample (bioactive peptides at known concentration) and A_b was the absorbance of sample blank (absence of enzyme). Besides, the bioactive peptides also could double stimulate the secretion of growth hormone and insulin, thus stimulating the absorption of glucose by muscle cells, preventing protein degradation, and achieving rapid repair of skeletal muscle.

Bioactive peptides relationships to oxidative stress and cellular metabolism in skeletal muscle

Skeletal muscle-induced oxidative stress is normally unavoidable, which is caused by the imbalance between production of reactive oxygen species and the elimination of antioxidant

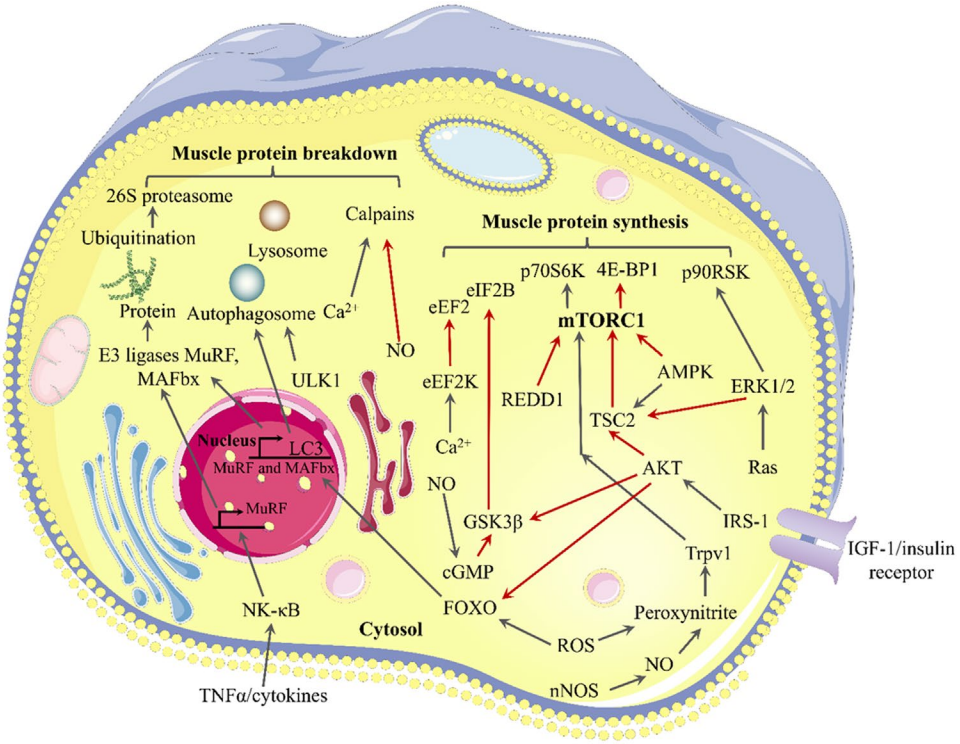

Figure 1. Mechanism diagram of key signal pathways of protein synthesis and degradation in skeletal muscle. The black arrows represented stimulus signals, while the red arrows represented suppression signals. 4E-BP1: eukaryotic initiation factor 4E binding protein; AKT: protein kinase B; AMPK: AMP-activated protein kinase; cGMP: cyclic guanosine monophosphate; c-Myc: c-myelocytomatosis oncogene; eEF2: eukaryotic elongation factor 2; eEF2K: eukaryotic elongation factor 2 kinase; eIF2B: eukaryoticinitiation factor 2B; eIF3f: eukaryotic initiation factor 3f; ERK1/2: extracellular signal-regulated kinase 1/2; FOXO: forkhead box O protein; GSK3β: glycogen synthase kinase 3β; IGF-I: insulin-like growth factor 1; IRS-I: insulin receptor substrate 1; LC3: microtubule-associated proteins 1 A/1B light chain 3B; MAFbx: muscle atrophy F-box; Mtorc1: mammalian/mechanistic target of rapamycin complex 1; MuRF1: muscle RING finger; Nf-κB: nuclear factor κB; nNOS: neuronal NO synthase; NO: nitric oxide; p70S6K: ribosomal protein S6 kinase p70; p90RSK: ribosomal protein S6 kinase p90; REDD1: regulated in development and DNA damage response1; ROS: reactive oxygen species; TNFα: tumor necrosis factor alpha; Trpv1: transient receptor potential cation channel subfamily V member 1; TSC1/2: tuberous sclerosis complex1/2; ULK1: unc-51-like autophagy activating kinase.

mechanisms (Arazi, Eghbali, and Suzuki 2021; Jia and Di 2023). Reactive oxygen species, as the core factor of signal molecules regulating cell and physiological function, plays a key role in exercise-induced pathophysiology of skeletal muscle, which occurs in the exercise-dependent manner. The production of reactive oxygen species increases with the increase of skeletal muscle contraction activity, which leads to numerous physiological and biochemical adaptation including mitochondrial biogenesis and changes in active muscle fibers (Lv et al. 2022; Jia, Di, et al. 2023). A previous study found that skeletal muscle injury and subsequent repair were accompanied by the recruitment of macrophages and phagocytes cells, which were the additional sources of reactive oxygen species production, causing injury of skeletal muscle tissue itself and skeletal muscle impairment (Thirupathi et al. 2020). Exercise-induced oxidative stress is also related to the expression level of endogenous antioxidants, which cannot surmount all the damage associated with oxidative stress in cells especially under physiological stress. Consequently, skeletal muscle-induced oxidative stress requires therapeutic intervention, such as dietary supplements containing antioxidants to regulate the redox process for maintaining muscle performance. Fan et al. (2022) investigated the moderation effects of chicken-derived bioactive peptides Leu-Lys-Tyr, Val-Arg-Pro, Val-Arg-Tyr and Val-Val-His-Pro-Lys-Glu-Ser-Phe on the tumor necrosis factor alpha (TNFα) induced oxidative stress in human vascular endothelial cells. Among the four peptides, Val-Val-His-Pro-Lys-Glu-Ser-Phe could activate the expression of antioxidant enzymes in cells that terminated reactive oxygen species, which significantly inhibited the increase of NADPH oxidase activities. In addition, Val-Val-His-Pro-Lys-Glu-Ser-Phe treatment reduced the cytoplasmic membrane translocation of $p67^{phox}$, which was the main reactive oxygen species generation and source of a subunit of NADPH oxidase 2. Therefore, the major mechanism of Val-Val-His-Pro-Lys-Glu-Ser-Phe antioxidation might be the inhibition of NADPH oxidase activity or the reduction of TNFR1 expression on the cell membrane linked to NADPH 2 by through Val-Val-His-Pro-Lys-Glu-Ser-Phe treatment. The protective effects of Val-Val-His-Pro-Lys-Glu-Ser-Phe on oxidative stress had laid foundation for potential therapeutic effect on diseases. The soft-shelled turtle peptides were applied to investigate the antifatigue effects on mice, which also could activate the NRF2/KEAP1 signaling pathway (Zhong et al. 2022). NRF2 had the function of regulating the expression of 4 main genes involved in NADPH production, which was a pivotal cofactor supporting the antioxidant process. Soft-shelled turtle peptides enhanced the antioxidant capacity of body by providing hydrogen atoms or electrons, which contributed to chelate metal ions, eliminate free radical reactions, inhibit the production of reactive oxygen species and up regulate antioxidant enzymes. Soft-shelled turtle peptides supplement successfully increased the activity of superoxide dismutase and glutathione peroxidase while decreasing the level of malondialdehyde, which confirmed that soft-shelled turtle peptides showed antioxidant capacity by maintaining the balance between antioxidant and oxidation mechanisms. The study proved that soft-shelled turtle peptides supplement had the ability of enhancing in exercise endurance, increasing energy storage and antioxidant activity, which also contributed to confirm this beneficial effect in the human body.

Bioactive peptides relationships to inflammation in skeletal muscle

Intense exercise typically causes skeletal muscle damage and inflammatory reaction, which play a vital role in the regeneration process. After skeletal muscle injury, blood monocytes rapidly invaded the damaged tissue as well as differentiated into $Ly6C^{high/pos}$ pro-inflammatory injury related macrophages firstly (Figure 2). They expressed that pro-inflammatory effectors including interleukin (IL)-1β or tumor necrosis factor can support the increase of inflammatory reaction, and contribute to the proliferation of MUSC while limiting the expansion of FAP. In addition, they also participated in the phagocytosis or effervescence of cell fragments to trigger their inflammatory transformation, which was the transition from pro-inflammatory $Ly6C^{high/pos}$ injury related type to the anti-inflammatory $Ly6C^{low/neg}$ restorative phenotype (Juban 2021). Food derived proteins and food derived bioactive peptides, as the treasure trove for potential natural therapies, are widely applied to study the anti-inflammatory potential. Chakrabarti et al. (2017) analyzed the anti-inflammatory ability of milk derived bioactive peptides Val-Pro-Pro and Ile-Pro-Pro based on mammalian cell culture experiments. Angiotensin II mediated up regulation of COX-2 (a pivotal pro-inflammatory mediator), was blocked by Val-Pro-Pro, demonstrating the potential as an inflammatory environment regulator. Intriguingly, Ile-Pro-Pro had no consistent effects on the parameters, which showed different reactions to inflammatory changes. Angiotensin II activated a great deal of pro-inflammatory pathways in the vascular wall, including NF-κB pathways and MAP kinase (such as ERK1/2), both of which could promote proliferation and inflammation of cells. Val-Pro-Pro and Ile-Pro-Pro were regulated to inhibit angiotensin type 1 receptor activation and/or the level of signal cascade downstream of the receptor. Val-Pro-Pro could attenuate the activation of p65, a component of NF-κB pathway, which was partly involved in the down-regulation of COX2 expression to regulate inflammation. The role and mechanism of milk derived bioactive peptides Val-Pro-Pro and Ile-Pro-Pro in the anti-inflammatory and antioxidant pathways provided a theoretical basis for subsequent researches. In addition, collagen peptides may also have anti-inflammatory properties due to the inhibitory effect of glycine on the release of proinflammatory cytokines (such as interleukin-6) (Khatri et al. 2021). The beneficial impact of collagen peptides might be due to the activation of undenatured type II collagen specific T regulatory cells. Type II collagen could release anti-inflammatory cytokines (transforming growth factor-β and interleukin-10), which might antagonize the inflammatory cascade reaction related to strenuous exercise, thus leading to the transformation of chondrocytes to extracellular matrix supplementation. Therefore, collagen peptide accelerated the recovery of joint injury and improved joint function through the anti-inflammatory properties or influence on extracellular

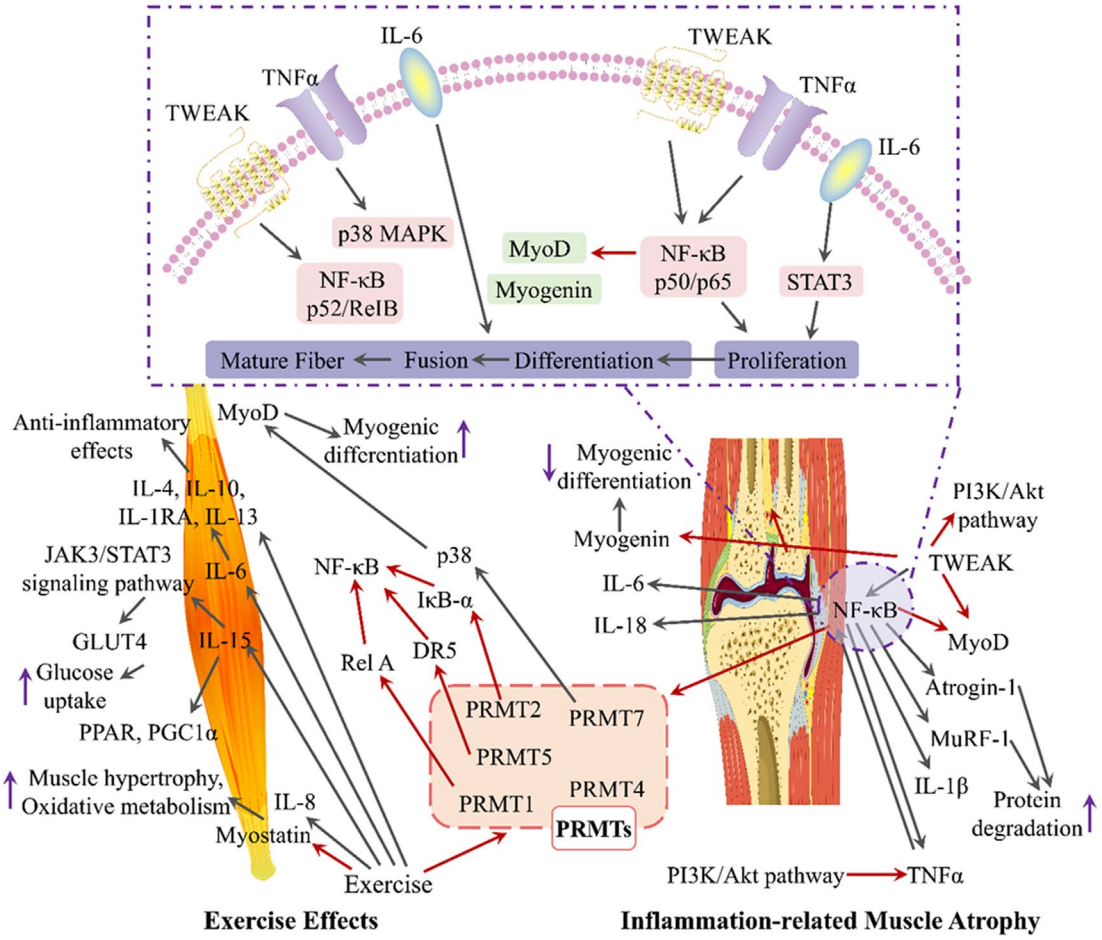

Figure 2. Mechanism diagram of protein arginine methyltransferase in inflammation related muscle atrophy and exercise mediated muscle metabolism and hypertrophy. Protein arginine methyltransferase could regulate the activities of Nf-κB va multiple mechanisms to fine-tune the inflammatory responses caused by pro-inflammatory and anti-inflammatory cytokines during exercise. IL-10: interleukin 10; IL-13: interleukin 13; IL-15: interleukin 15; IL-1RA: interleukin 1 receptor antagonist; IL-1β: interleukin 1β; IL-4: interleukin 4; IL-6: interleukin 6; IL-8: interleukin 8; MuRF-1: Muscle RING-finger containing protein-1; Nf-κB: nuclear factor κB; PGC1α: proliferator-activated receptor gamma coactivator-1alpha; PRMT: protein arginine methyltransferase; STAT3: signal transducer and activator of transcription 3; TNFα: tumor necrosis factor alpha; TWEAK: tumor necrosis factor-like weak inducer of apoptosis.

matrix regeneration and collagen synthesis in cartilage and tendon. Oertzen-Hagemann et al. (2019) investigated the effects of collagen peptide with resistance exercise training supplementation on the protein composition of skeletal muscle by using proteomics. Collagen peptides induced a higher increase in the proteins (including actin-binding proteins, myosin proteins and tropomyosin) associated with resistance training adaptation, which might be due to the previously observed higher concentration of hydroxyproline peptide in collagen peptide supplements. Despite these evidences, how the anti-inflammatory effects of these bioactive peptides contribute to better muscle recovery after exercise-induced skeletal muscle damage in humans still needs further research.

Bioactive peptides relationships to cartilage and functional joint pain in skeletal muscle

People with resistance training may be more likely to show the risk of functional joint pain and posttraumatic osteoarthritis, which is caused by the imbalance in anabolic and catabolic processes in articular cartilage and the changes of proinflammatory throughout the joint (Zhao et al. 2019; Jia, Du, et al. 2023). Some evidences showed that oral administration of specific collagen peptides such as skin collagen peptides, gelatin hydrolysate peptides, collagen tripeptide combined with injections and hyaluronic acid could reduce functional joint pain and posttraumatic osteoarthritis (Figure 3). The potential mechanisms of collagen peptide in relieving functional joint pain and posttraumatic osteoarthritis were as follows: (*i*) promoting the synthesis of type II collagen in chondrocytes; (*ii*) inhibiting the denaturation of type II collagen and the expression of matrix metalloproteinase-13; (*iii*) reducing cartilage inflammation that was related to the reduction of cartilage oligomeric matrix protein, tumor necrosis factor-α and interleukin-1 (Cui et al. 2022b).

Cook Sangar et al. (2020) investigated the cystine-dense peptides and the effects on the cartilage of ankles, knees, hips, intervertebral disk and shoulders. The cystine-dense peptides-11R could reach the peak concentration in the cartilage of ankles, knees, hips, intervertebral disk and shoulders within 30 min after the administration and were still

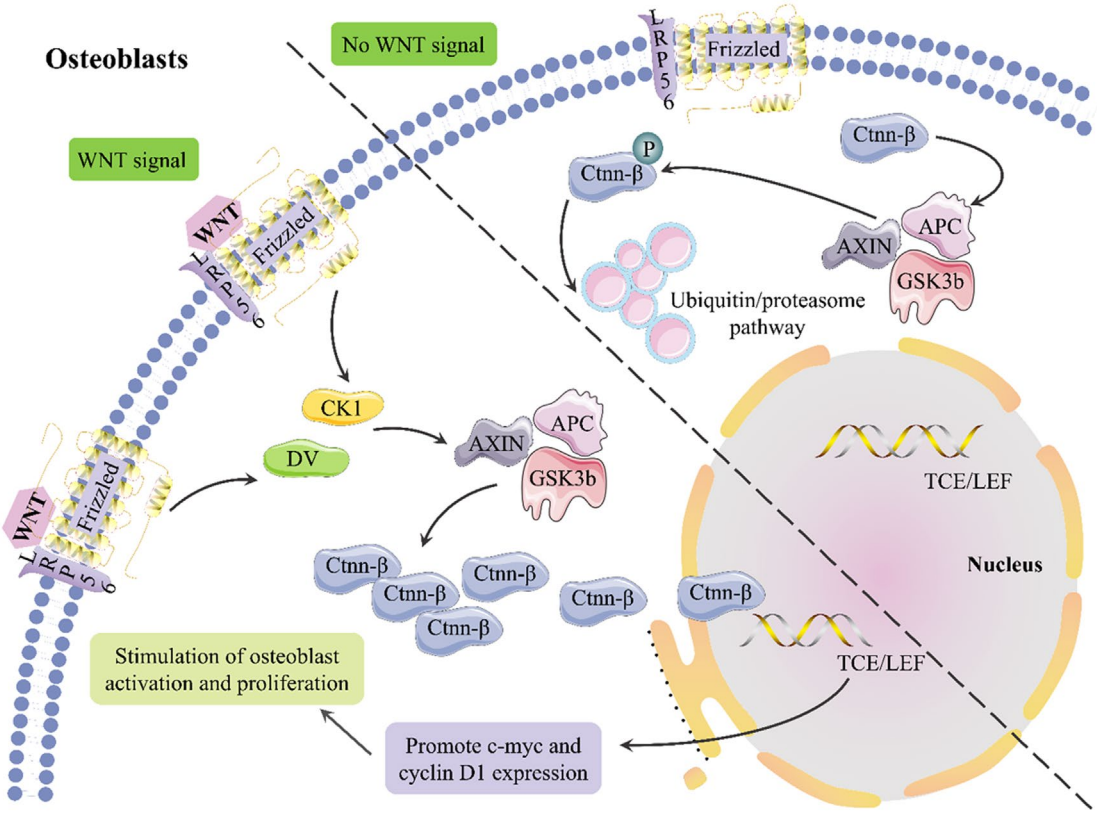

Figure 3. Pathway map of osteoblast activation and proliferation regulation. Collagen peptide promoted osteoblast differentiation and calcium absorption by activating PI3K/Akt pathway, as well as then activated Wnt/β-catenin signaling pathway in p38MAPK signal pathway. Wnt/β-catenin signaling pathway regulated the activation and proliferation of osteoblasts.

detectable for more than 4 days. Cystine-dense peptides-11R with one of two different steroid payloads (triamcinolone acetonide and dexamethasone) or a fluorophore imaging agent could distribute in the joints. Of the two different steroid payloads, the dexamethasone conjugate did not develop due to the free drug released into the circulation leading to targeted toxicity. On the contrary, triamcinolone acetonide conjugate alleviated the arthritis in the rat collagen induced rheumatoid arthritis model, while avoiding the toxicity of untargeted steroid therapy at the same molar dose. The research provided laid a foundation for the multi joint targeting of disease modification therapy payloads and might create opportunities for meaningful improvements in human health. Furthermore, key bioscience peptides, as the synthetic peptides based on salmon calcitonin, were applied to explore the mechanism of inhibiting bone resorption and chondroprotective. Specific collagen peptides (TENDOFORTE®, GELITAAG) were taken by participants to investigate whether oral supplementation of specific collagen peptides improved symptoms and tendon vascularization in patients with chronic mid-portion (Praet et al. 2019). According to the Victorian Institute of Sports Assessment-Achilles questionnaires and contrast-enhanced ultrasound results, we could know that collagen peptides accelerated the clinical benefits of a well-structured calf-strengthening and return-to-running program in Achilles tendinopathy patients. In addition, bioactive peptides yielded improvements in muscle mass and strength during postoperative rehabilitation, which further reduced pain, increased physical mobility, and improved global functioning after 2–7 wk of rehabilitative exercise training. The results facilitated the formulation of practical and interdisciplinary approaches to cartilage and functional joint pain in skeletal muscle.

Bioactive peptides relationships to tendon properties of connective tissue in skeletal muscle

Tendons are mainly composed of collagen, which is layered in the form of fibrils, fibers as well as bundles. Collagen fibers, as the basic unit of tendons, account for more than 80% of the extracellular matrix (dry weight) in tendons and form a multilayer structure fibril network containing collagen fibers (Ellingson, Pancheri, and Schiele 2022). Most collagen of healthy tendons is type I collagen (Col-I) fibril, which has a stiff structure and provides mechanical strength and durability. Type II collagen is usually concentrated near the tendon bone insertion point and only exists in a small amount (Figure 4). The sliding and stretching of collagen structures play a momentous role in in the ability of tendons to transmit force between bone and muscle as well as conclusively form a significant contribution to the strength and deformation mechanism of tendons. Type III collagen fibers play a key role in the healing process, which are abundant in the scar tissue after tendon injury. Type III collagen fiber is thinner than type I collagen fiber, forming elastic and

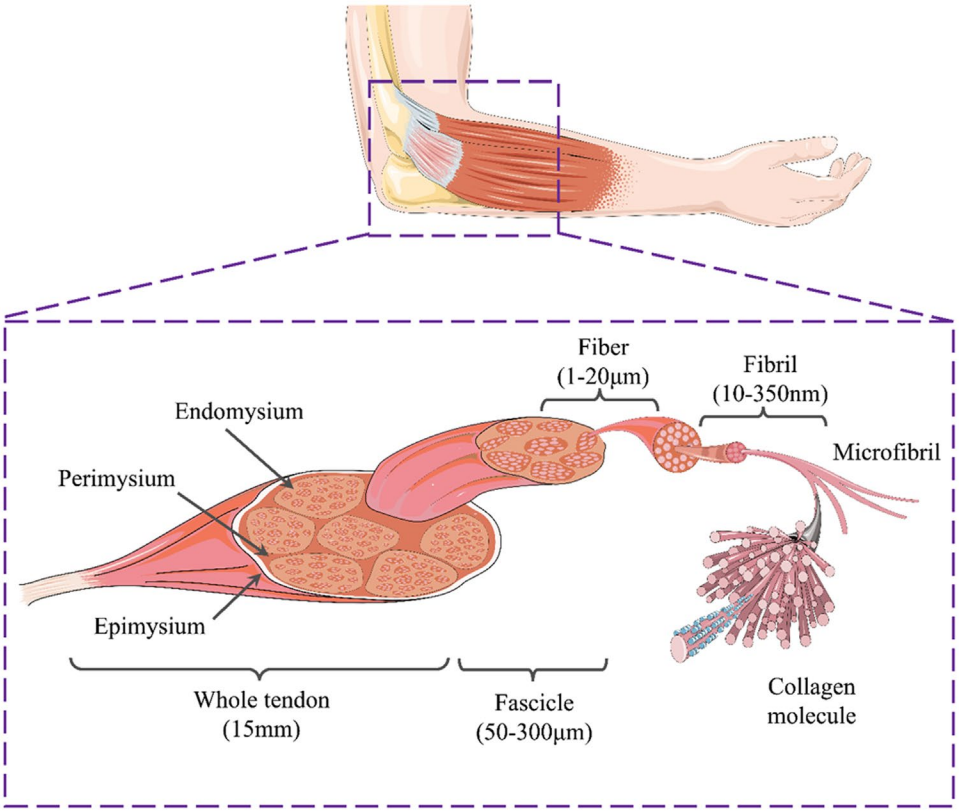

Figure 4. The tendon structure at different levels of tendon hierarchy.

loose tissue. The increased proportion of type III/I collagen in the tendon indicates that it is tendon degeneration and the cause of mechanical injury. After high-load resistance training, tendons are stimulated to produce extracellular matrix proteins, which can regulate the formation, maintenance as well as differentiation of myosatellite cells to enhance the adaptive response of tendons (Jia, Guo, et al. 2023). The anabolic processes could be positively affected by the application of collagen peptides, which could stimulate tenocytes to synthesize extracellular matrix proteins such as type I and III collagen and elastin. Collagen peptide was characterized by the low molecular weight as well as high proportion of and hydroxyproline and proline, resulting in the higher transport efficiency and high resistance to intestinal digestion as well as consequently to a high bioavailability. Based on the evidence provided by the initial clinical studies, we could know that the combination of collagen peptide and training intervention could promote the rehabilitation of injured tendons. To understand the beneficial effects on mechanical properties and structural of the healthy Achilles tendon, Jerger et al. (2022) explored the effects of specific collagen peptide combined with resistance training on the changes of muscular and tendinous properties. A total of 40 healthy male volunteers (age: 26.3 ± 4.0 years) were asked to complete the 14 wk of heavy resistance training program. One group received 5 g of a placebo supplement, while the other group received a daily dosage of 5 g specific collagen peptides. Changes in cross-sectional area of Achilles tendon, muscle strength, thickness of the plantar flexors and tendon hardness were detected. The positive influences of collagen peptide supplementation on the muscle hypertrophy might be due to the increased phosphorylation of muscle proteins synthesis markers (including p70S6K, mTOR, Akt) and incremental contractile element proteins. The results of the primary endpoint showed that supplementing specific collagen peptides combined with resistance training was associated with more hypertrophy in the structures of tendons and muscle than resistance training alone in the young physically active men. Therefore, oral administration of specific collagen peptides might potentially benefit to the rehabilitation and prevention of tendon injuries.

Dietary sources of bioactive peptides

Food-derived bioactive peptides, as functional peptides extracted indirectly or directly from food proteins by organisms *in vitro*, commonly exist in dietary proteins with the particular amino acid sequences (Yang et al. 2021a; Ye et al. 2023). Up to now, food-derived bioactive peptides have been detected in meat, fish, milk, egg, vegetables, fruits and among others, which were discussed here. In the section that follows, the production of bioactive peptides from natural sources, the structure activity relationship, the functional properties and the functional mechanisms of food-derived bioactive peptides from all kinds of food raw materials were summarized and analyzed.

Influence of meat-derived bioactive peptides on the treatment of skeletal muscle injury

Meat and meat products, as the source of health-promoting peptides containing a great quantity of essential amino acids, were superior to other protein sources in the production of low molecular weight food-derived bioactive peptides due to the high percentage of protein and availability (Wang et al. 2021b; Manzoor, Singh, and Gani 2022). Meat-derived bioactive peptides were hidden in the primary structure of proteins but could be affected by digestive enzymes *in vitro* or in the intestine, which could pass through the intestine and enter the circulation, in which they could play its roles in various functions. The bioactivity of peptides derived from meat protein was related to the terminal amino acid type, amino acid sequence, terminal amino acid charge, molecular weight, spatial structure, hydrophobicity and hydrophilicity, which exhibited the inhibitory and antioxidant characteristics of angiotensin converting enzyme. On the scale of the laboratory, meat-derived bioactive peptides were generated by alkali, acid or enzymatic hydrolysis, which were followed by the fractionation and purification processes to produce concentrated peptide. Up to now, the majority of bioactive peptides were discovered from various types of meat such as beef, duck and mutton, pork and chicken (Madhu et al. 2022). Frozen beef and fish samples were processed to purify bioactive peptide, which were evaluated in its angiotensin converting enzyme inhibitory properties, antioxidant and antibacterial (Maky and Zendo 2021). After pepsin hydrolysis for 8 h, Sep-Pak C18 cartridge and reversed-phase high-performance liquid chromatography were used for purification. On the basis of Edman degradation, there were 34 amino acid residues to form beef peptide and 21 amino acid residues to make up fish peptide. Beef peptide had more hydrophobic residues at N-terminal (including Ile, Ala and Leu) with higher angiotensin converting enzyme inhibitory activity. The different amino acid sequence from SJGAP and YFGAP were identified from fish peptide, which was highly similar with GAPDH and prevented the cell death. Bioactive peptides with antioxidant potential and cytoprotective effects on oxidative stress induced by hydrogen peroxide were isolated and identified in duck. A total of 6 novel peptides (DTYIRQPW, GEHGDSSVPVWSGVN, HLDYYLGK, HLTPWIGK, MYPGIAD and WDDMEKIWHH) were detected, which had hydrogen bonds and hydrophobic amino acids, resulting in antioxidant peptides with radicals-scavenging activities. The protective effects of these 6 antioxidant peptides on H_2O_2-induced oxidative damage in HepG2 cells were mainly related to the enhancement of superoxide dismutase and catalase activities and the inhibition of lipid peroxidation (Sun et al. 2022; Xu et al. 2021). Wang et al. (2020) discovered 3 antioxidant peptides (Ala-Pro-Tyr-Met-Met, Met-Trp-Thr-Asp and Phe-Trp-Ile-Ile-Glu) from mutton ham, which had the ability to inhibit Fenton's reagent-induced protein oxidation, lipid oxidation as well as DNA damage. Gonzalez-Noriega et al. (2022) collected peptide fractions and hydrolysates from the pork and chicken skin collagen extracts, which had a good inhibitory activity of pancreatic lipase *in vitro*. The pancreatic inhibitory peptides from pork and chicken had multiple proline and glycine residues in in their sequences, especially at their C- and N-terminal, which indicated that they were related to their biological activities and applied for obesity treatment. The chicken sternal cartilage hydrolysates were also applied for osteoarthritis treatment, which were mainly comprised of hyaluronic acid, collagen peptides and chondroitin sulfate. Ma et al. (2021) investigated the preventive influences of chicken sternal cartilage hydrolysates on osteoarthritis, which were induced by medial meniscectomy and anterior medial collateral ligament surgery in rats. Chicken sternal cartilage hydrolysates could significantly alleviate the cartilage damage through inhibiting chondrocyte changes, which also could destroy the collagen structure and improve anomalous collagen mechanism through inhibiting collagen degradation. The study provided new insights into clinical treatment of skeletal muscle injury with meat-derived bioactive peptides (Table 1).

Influence of marine organism -derived bioactive peptides on healing of skeletal muscle injuries

Fish, as a traditional food, is consumed by many cultures and contains several bioactive compounds including peptides, proteins, vitamins, ω-3 polyunsaturated fatty acids and minerals (Heffernan, Giblin, and O'Brien 2021). Enzymatic hydrolysis, fermentation and synthesis were the most popular approach to obtain bioactive peptides from fish (Figure 5). The production of bioactive peptides and softening of fish texture were caused by the hydrolysis of myofibrillar proteins, which was related to the activity of endogenous proteases (such as cathepsins and calpains). Endogenous proteolytic activity can produce bioactive peptides fragments, including amino acids, short peptides and oligopeptides. Cathepsin B with a potent dipeptidyl carboxypeptidase (peptidyl-dipeptidase) activity could generate dipeptides by the C-terminal cleavage of proteins and peptides, which were released in fish and utilized as food components (Cerrato et al. 2021; Jia, Ma, et al. 2023). According to the report, Fimbres-Romero et al. (2021) once exploited the hydrolysis of squid muscle collagen into peptides by the crude extracts of collagenases from Sierra fish (Scomberomorus Sierra) viscera and jumbo squid (Dosidicus gigas) hepatopancreas and verified proteins from 20 to 200 kDa. Collagen, as the major component in connective tissue, was composed of hydrophobic amino acids (Gly, Hyp and Pro). The Hyp-Gly-Pro sequences were common in collagen, which were momentous to enhance the biological activity of peptides produced by collagen hydrolysis due to the radical trapping capacity of Gly and Pro. Side chain of Gly was composed of a single hydrogen atom, which could endow the peptide flexibility. Zamorano-Apodaca et al. (2020) obtained 5 peptide fractions by the hydrolysis of acid-soluble collagen generated from various fish species to evaluate the functional (emulsifying ability, foaming and solubility) and biological (antimicrobial and antioxidant) activities. Therefore, the five peptide fractions might become the potential substitute for ingredients in the production of

Table 1. Alternative and unconventional and sources of bioactive peptides from food.

Classification	Source	Peptide	Peptide sequence	Biological activity	Key findings	Reference
Plant source	Coffee beans	Dipeptidyl peptidase-IV and angiotensin I-converting enzyme inhibitor peptides	GW, PL, GL, GF, VY, VF, VAF and VPK	Inhibitory activities (AH, AL, VY, GL and TY), free radical scavengers (GL and AL), antioxidant capacity	The green coffee protein extract showed lower antioxidant activity when compared to the roasted coffee bean and spent coffee ground protein extracts, which could be ascribed to the interactions occurring between melanoidins and the proteins during roasting.	(Ribeiro, Rocha, and Prudencio 2021)
	Pixian broad-bean paste	Taste peptides	ALDELGT, AELTPEP, SAALQAG, SFEAVEAAPT, EQDNDGNNIFSGFKR, QTFNTEEDTAK, DAPASGN, GGLRIINPEGQQ, SIITPPERQ, GGLSIITPPERQA, GGLRIINPEG, SIITPPER, RIINPEGQQ, DALNVNRHIV, and LPKILLLQLV	Umami as well as salty- and umami-enhancing peptides	The different taste of peptides was governed by the amino acid sequence rather than individual amino acids.	(Zhao et al. 2022)
	Rice	Antioxidant peptides	Gly-Asp-Met-Asn-Pro and Leu-Leu-Leu-Arg-Trp	Higher antioxidant activities	These peptides were synthesized and also exhibited good scavenging activity on the superoxide anion free radical, DPPH free radical and hydroxyl radical as well as good chelating ability on Fe^{2+}.	(Chen et al. 2021)
	Wheat	Wheat gluten bioactive peptides	WGLYH	Anti-diabetic activities	WGLYH peptide might contain anti-diabetic peptides and had potential for anti-diabetic applications.	(Mousavi, Azizi, and Abbasi 2022)
	Mung bean	Antihypertensive peptides	LPRL, YADLVE, LRLESF, HLNVVHEN, and PGSGCAGTDL	Inhibiting *in vitro* activities of angiotensin, I-converting enzyme and renin	MPH had the weakest antihypertensive activity. PGSGCAGTDL had the weakest persistence and indicated greater susceptibility to the endogenous proteolysis.	(Sonklin et al. 2020)
	Acacia catechu	Antiviral peptides	HVTPDIAYNPRTYM and DHVTPDIAYNPWAYF	Anti-dengue virus activity	Novel peptide inhibitors were identified from medicinal plants that dramatically inhibit dengue virus infection in the early steps of infection. Peptide inhibitors identified in Acacia catechu were found to inhibit all four dengue virus serotypes.	(Panya et al. 2019)
	Chia seed	Bioactive peptides of anti-inflammatory activity	NSPGPHDV ALDQ and RMVLPEYELLYE	Anti-inflammatory activity	Peptides prevented adipogenesis by modulating PPARγ and additionally, preventing and even inhibiting inflammation in adipocytes by inhibition of PPARγ and NF-κB expression.	(Grancieri, Martino, and Gonzalez de Mejia 2021)
	Corn	Immunomodulatory peptides	PFNQL, FLPFNQL, SQLALTNPT, GAPFNQL and FLPPVT	Antioxidant, anti-proliferative and anti-hypertensive activities	Zein protein hydrolysate digested by thermolyzing showed a high yield of peptides with potent activity.	(Liu et al. 2020)
Animal source	Cheddar cheese	Antioxidant peptides	VLPVPQK, EDVPSER, DKIHPF, EMPFPK, KEMPFPK and SDIPNPIGSENSEK	Antioxidant activity and angiotensin I-converting enzyme-inhibitory activity	The nutrient composition of cheese was not affected by the addition of Lactobacillus Helvetius. The antioxidant activity of cheese during ripening time could significantly increase.	(Yang et al. 2021b)
	Pig bone collagen	Calcium-binding peptide	AGPAGPAGAR, AGPHGPPGKDGR and GPAGPHGPPGKDGR	Promoting calcium absorption	Calcium could be chelated by carboxyl oxygen and amino nitrogen atoms of collagen peptides, thus forming peptide-calcium chelate.	(Wu et al. 2019)
	Atlantic sea cucumber	Antioxidant peptides	TEFHLL	Antioxidant activity	The alcalase-produced hydrolysates generally showed higher antioxidant activity than the trypsin-produced ones.	(Zhang et al. 2020b)
	Porcine liver	Antioxidant peptides	APAAIGPYSQAVLVDR, GLNQALVDLHALGSAR, ALFQDVQKPSQDEWGK and LSGPQAGLGEYLFER and LGEHNIDVLEGNEQFINAAK	Antioxidant capacity	The peptides most abundant and correlated with antioxidant capacity were APAAIGPYSQAVLVDR from uncharacterized protein, GLNQALVDLHALGSAR, ALFQDVQKPSQDEWGK and LSGPQAGLGEYLFER from ferritin and LGEHNIDVLEGNEQFINAAK from trypsinogen.	(Lopez-Pedrouso et al. 2020)
	Buffaloe skim milk	Angiotensin I-converting enzyme-inhibitory and antioxidant peptides.	FPGPIPK, IPPK, IVPN, QPPQ, YPSG, HPFA and KFQ	Angiotensin I-converting enzyme inhibitory activity and antioxidant capacity	Hydrolysis of buffaloes' skim milk retentate by papain, pepsin or trypsin enhanced the angiotensin I-converting enzyme-inhibitory activity and radical scavenging capacity. The papain hydrolysate showed the highest angiotensin I-converting enzyme-inhibitory activity and radical scavenging capacity.	(Abdel-Hamid et al. 2017)
Aquatic source	Arthrospira (Spirulina) platensis	Antioxidant peptide	GMCCSR	Anti-oxidant activity	The peptide could protect erythrocytes from lipid peroxidation and protein oxidation.	(Zeng et al. 2018)
	Arthrospira platensis	Antioxidant peptide and angiotensin I-converting enzyme inhibitory peptides	VTAGLVGGGAGK and PTGNPLSP	Radical scavenging activity, hydroxyl radical scavenging activity, ferrous ion chelation activity and ACE inhibitory activity	The PTGNPLSP was more stable than VTAGLVGGGAGK during *in vitro* gastrointestinal digestion.	(Wang et al. 2021a)
	Black sharkskin	Antioxidant peptide	ATVY	Anti-oxidant activity	The black sharkskin-derived novel antioxidant peptide (ATVY) demonstrating 81.05% free radical scavenging activity to ABTS.	(Yang et al. 2020)

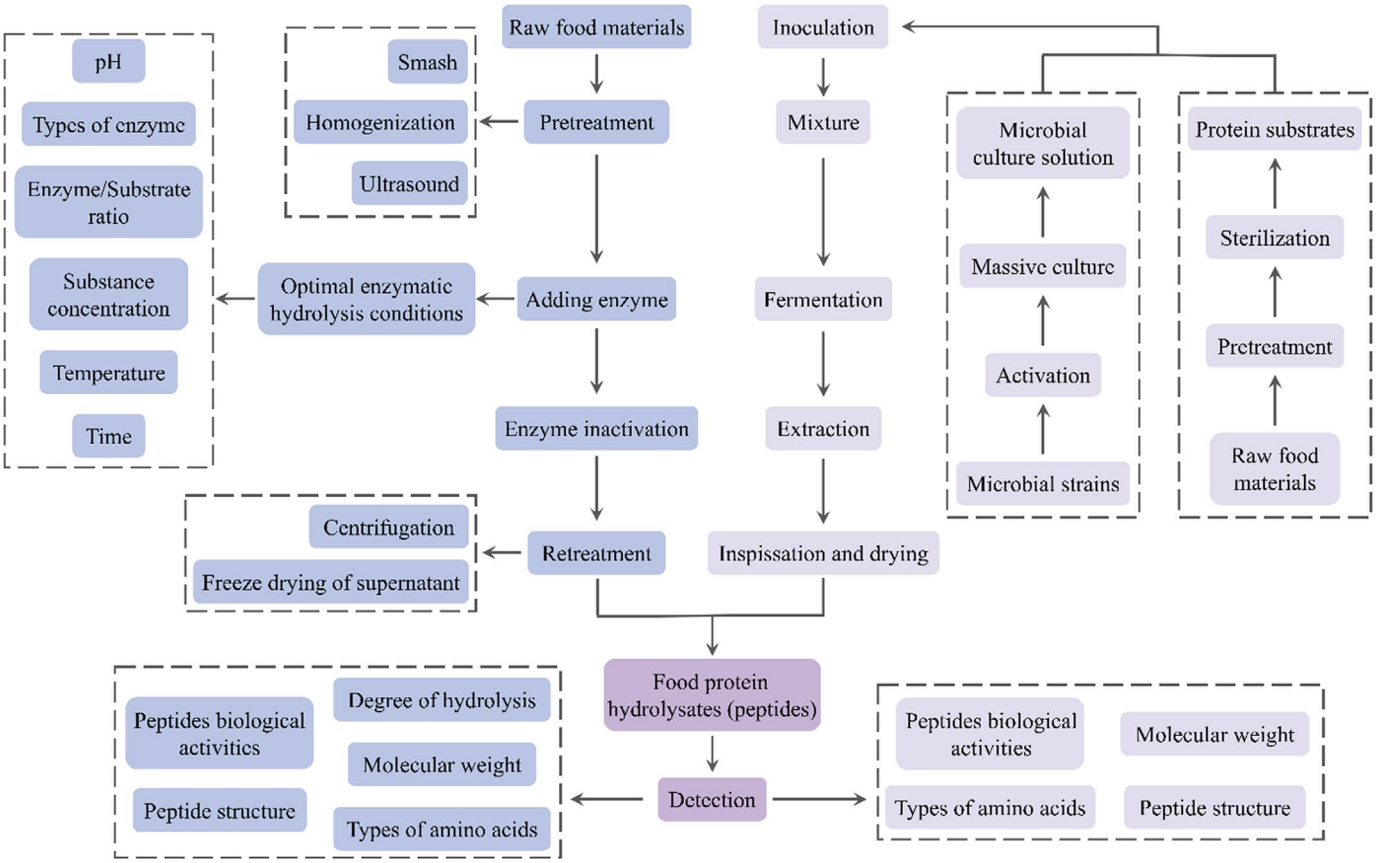

Figure 5. The steps in the enzymatic and microbial fermentation preparation of bioactive peptides.

functional food. With the development of fish-derived bioactive peptides separation and purification techniques, an increasing number of peptides are used in clinical investigation. Marine fish peptides (collagen peptides) compound was applied to explore the wound healing potential in rats after cesarean section by biomechanical, biochemical and histological methods (Peng et al. 2020). In marine fish peptides group, the formation of smooth muscle fibers and the expression levels of CD34 and connective tissue growth factor were flourishing. CD34 could accelerate the migration of vascular endothelial cells with the synergism of chemokines and adhesion molecules, which was instrumental in vascular reconstruction and endothelial repair. The increased expression of connective tissue growth factor promoted collagen synthesis, collagen deposition, fibroblast differentiation and proliferation for wound repair. Furthermore, marine fish peptides had a large content of Hyp, which was the primary component of collagen fibers in the connective tissue and the level of marine fish peptides represented the amount of granulation tissue and collagen formation. Hyp content, as a sign of collagen deposition, had a great impact on the tensile strength of connective tissue, which was necessary for the healing of skeletal muscle injuries in animals and humans (Gao et al. 2021). Hyp was conducive to the collagen molecules stability, collagen synthesis, tensile strength as well as tissue repair, which played a key role in the formation and stability of triple helix of collagen molecules at suitable temperature.

Influence of dairy products-derived bioactive peptides on skeletal muscle protein anabolism

Dairy products are considered as the "ultimate" food, which can provide immunological protection, critical nutritive elements and biologically active substances to people and ensure growth and safe development during life (Jia and Wang 2023). The dairy products-derived bioactive peptides were first detected in 1950, which can remain inactive or latent in the precursor protein sequence until released in various ways (Wang et al. 2022a). Among them, microbial fermentation, plant and bacterial proteases as well as hydrolysis by digestive enzymes are the most familiar approach of bioactive peptides release (Ryan et al. 2011). According to the report, the bioactive peptides also were formed during the gastrointestinal digestion, which was beneficial for improving the biological activity of peptides. Whey proteins and casein derived from bovine milk, as the source for bioactive peptide, had been applied to screen peptide fragments during *Lactobacillus brevis* CGMCC15954, *Lactobacillus plantarum* A3 and *Lactobacillus reuteri* WQ-Y1 fermentation process (Cui et al. 2022a). Some antioxidant peptides sequences in fermentation processing were confirmed, including VKEAMAPK (β-casein), YLGYLEQLLR (αS1-casein) and YIPIQYVLSR (κ-casein), which had the potential antioxidant capacity due to the repetitious amino acids and lysine at the C-terminal of bioactive peptide sequences. Furthermore, the sheep cheese whey was

hydrolyzed by novel fungal and bacterial food-grade protease preparations. The bacterial was more suitable than fungal for hydrolyzing sheep cheese whey to produce angiotensin-I converting enzyme inhibitor bioactive peptides and antioxidant (Welsh et al. 2017). Li et al. (2020) investigated the influence of α_{s1}-casein polymorphism on digestibility and biological activity of buffalo milk protein in during simulated gastrointestinal digestion. The amino acid substitution in α_{s1}-casein polymorphism changed the bioactive peptides releasing behavior and affected the digestibility through the sensitivity of cleavage site of the variant to pepsin and pancreatin. A total of 3 bioactive peptides (including FFVAPFPEVFGK, HIQKEDVPSER and YLGYLEQLLR) were identified in two variants of αs1-casein and jejunum of healthy humans, which could exert antioxidant and anxiolytic activities. At the current study, ten Haaf et al. (2021) compared the carbohydrate supplementation and iso-caloric protein supplementation (40 g milk protein (20% whey protein, 80% casein) prior to sleep and 20 g milk protein during breakfast and after exercise) on the muscle damage or exercise-induced muscle soreness. The optimized protein supplement could reduce muscle damage and induce the repair as well as growth of contractile protein through active muscle protein balance (Jia, Wang, et al. 2023). Carbohydrate supplementation and habitual protein intake on muscle protein balance had independent and synergistic effects in reducing muscle soreness. Whey protein, as the mixture of proteins isolated from cheese (including β-lactoglobulin, glycomacropeptide, lactoferrin, immunoglobulins and α-lactalbumin), can be rapidly digested into free amino acids and absorbed in plasma to induce skeletal muscle protein anabolism (Han et al. 2022). Shin et al. (2020) utilized four diverse whey protein hydrolysates to administer C57BL/6 mice with muscle atrophy for 2 wk. Research showed that whey protein hydrolysate inhibited the PI3K/Akt/forkhead box O pathway and conspicuously promoted the phosphoinositide 3-kinase/protein kinase B/mammalian target of rapamycin pathway, which provided a superior basis for developing functional food to prevent the sarcopenia.

Influence of egg-derived bioactive peptides on protein synthesis in skeletal muscle

Eggs, as an affordable nutrient-rich food, contain a great deal of digestible proteins, bioactive peptides, lipids, carbohydrates, vitamins and a trace of minerals, which are considered one of the most-consumed food and play an essential role in diet (Moreno-Fernández, Garcés-Rimón, and Miguel 2020). Egg proteins (such as ovotransferrin, ovalbumin, lysozyme and phosvitin) had been become a momentous source to widely generate novel bioactive peptides in last years (Benede and Molina 2020; Jia, Wu, and Kang 2023). Egg-derived bioactive peptides were obtained from the hydrolysis with proteolytic enzymes from bacterial, plant or animal origin such as alcalase, papain, pepsin and trypsin. In order to improve enzymatic digestibility, egg proteins will be modified by several physical methods including heat and high-pressure treatments, high-intensity pulsed electric field and sonication, so as to increase the free radical scavenging capacity of protein conjugate (Dong et al. 2019). Once the egg proteins hydrolysate is obtained, it can be separated into multifarious fractions by employing liquid chromatography with ion exchange, gel permeation as well as immobilized ligand affinity chromatography for better evaluation and determination of the functional properties. Mohd Salim and Gan (2020) established a comprehensive bioinformatics method of identifying dipeptidyl peptidase-4 and angiotensin converting enzyme inhibitory peptides from the egg white ovalbumin. After egg white ovalbumin was in silico digestion through various enzymes or enzyme combinations, a total of 10 highly potential peptides (including ADHPF, AM, CF, ELPF, FR, GR, KM, LPR, PR and PRM) were screened for *in vitro* validation experiment, which could inhibit dipeptidyl peptidase-4 and angiotensin converting enzyme to reduce the aggravation of hypertension and diabetes. For exploring more bifunctional egg-derived bioactive peptides, Zhang et al. (2020a) optimized the enzymatic hydrolysis process of egg white protein and purified the egg-derived bioactive peptides. Four peptides (Ala-Asp-Trp-Ala-Lys, Leu-Ala-Pro-Tyr-Lys, Pro-Lys-Ser-Val-Ile-Arg-Trp and Ser-Val-Ile-Arg-Trp) were discovered from fraction with highest bioactivity, which matched ovotransferrin as well as had alanine and leucine at N-terminus causing antioxidant. The angiotensin converting enzyme inhibitory mechanism of the four peptides were explored by molecular docking, which suggested that angiotensin converting enzyme inhibitory activity of Leu-Ala-Pro-Tyr-Lys was mainly caused by intermolecular interaction between Glu384 and Glu411. In addition, the egg white protein ovotransferrin-derived bioactive peptides Ile-Arg-Trp was applied to further explore the structure as well as activity relationships of Ile-Arg-Trp in tumor necrosis factor alpha induced insulin resistance L6 skeletal muscle cells (Jiang and Wu 2022). The tumor necrosis factor alpha treatment could impair insulin signaling pathway, glucose transporter protein 4 translocation and reduce insulin stimulated glucose uptake, which was related to the N-terminal Ile, C-terminal Trp and Arg with a positively charged. The structural requirements of Ile-Arg-Trp and the significance of each amino acid residue had been known in detail, which would provide constructive suggestions for future clinical treatment. A screening assay using C2C12 mouse muscle cells was developed to assess the relative abilities of egg-derived bioactive peptides, after simulated gut digestion, to activate the mechanistic target of rapamycin complex I (mTORC1) muscle protein synthesis signaling pathway (Roeseler et al. 2017). Egg-derived bioactive peptides could function as key stimulators of MPS *via* amino acid-sensitive activation of the mTORC1 pathway and its downstream target proteins, which could attenuate muscle wasting. To conclude, preclinical and human clinical trials would be essential to confirm and translate the results of such investigations to practical dietary recommendations and practices.

Influence of vegetable-derived bioactive peptides on oxidative stress in skeletal muscle

Vegetables are also easily accessible and affordable sources of proteins, which are packed with water, carbohydrates, dietary fibers, vitamins, minerals, alkaloids, carotenoids, phenolics, phytosterols, organosulfur compounds and nitrogen-containing compounds (Sosalagere, Adesegun Kehinde, and Sharma 2022; Jia, Wu, Liu, et al. 2023). Although the amount of protein hydrolysates and bioactive peptides from vegetables is not a majority, vegetable-derived bioactive peptides have been isolated from potato, spinach, algae, carrot, soy and others, as well as the exquisite bio-functional effects have also been investigated (Montesano et al. 2020). Earlier research results showed that potato protein hydrolysates exhibited the strong antioxidant activity and lipolysis-stimulating activity, which derived from alcalase treatment and had potential as effective anti-obesity dietary ingredient to prevent hepatosteatosis. Based on this, the regulatory effects of pure peptide amino acid sequence Ile-Phe and Asp-Ile-Lys-Thr-Asn-Lys-Pro-Val-Ile-Phe on cardiac apoptosis in spontaneously hypertensive rats had been investigated (Lin et al. 2020). In spontaneously hypertensive rats, Ile-Phe and Asp-Ile-Lys-Thr-Asn-Lys-Pro-Va l-Ile-Phe weakened cardiac apoptosis by upregulation of survival protein (PI3K/Akt), autophagy markers (Beclin 1 and LC3B) and mitochondrial biogenetic pathway proteins such as p-AMPKα, CREB, p-Foxo3a, Nrf2, PGC-1α and SIRT1. Scientific investigations had shown that spinach contained hormone-like peptides (insulin-like peptide and somatostatin-like material), which was confirmed by radioimmunoassay technology. Yesiltas et al. (2022) investigated 35 bioactive peptides derived from spinach, seaweed, potato and microbial proteins through the combination of bioinformatic prediction and quantitative proteomics. Numerous selected peptides such as YWTMWK from spinach protein, MLWQYKPK from microbial biomass, YKLLHCPSHLQCKN and DDDNLVLPEVYDQD from potato protein had the ability to significantly inhibit lipid oxidation, which provided valuable supplements to the public data set of antioxidant peptides. Carrot, as a valuable vegetable, is rich in carotene, vitamins, coumarin, volatile oil and others, which can inhibit the activity of amylase and absorb glucose as well as have good antioxidant capacity. Hao et al. (2022) purified a hypothetical carrots protein called Daucus Carrot hypoglycemic peptide, which had strong inhibitory activity against the α-glycosidase. It was found that Gln49, Glu43, Pro46, Thr47 and Thr48 were the key residues in Daucus Carrot hypoglycemic peptide to inhibit α-glycosidase activity through molecular docking and site-directed mutation. In order to prevent and treat type-2 diabetes mellitus, the investigation provided potential carrot-derived bioactive peptides as functional foods or nutraceutical supplements. Furthermore, soy proteins are also highly digestible and contain a great deal of essential amino acids, hydrolyzed and released by protein hydrolysis, gastrointestinal digestion and fermentation with starter cultures (Ashaolu 2020). Tonolo et al. (2020) extracted and identified bioactive peptides with potential antioxidant activity released by fermented soy products *in vitro* simulated INFOGEST protocol including 3 sequential phases: oral, gastric as well as intestinal. The soy-derived bioactive peptides had positive effects of on antioxidation, improvement of cell viability and ROS production, which also could monitor the Keap1/Nrf2 pathway and interact with amino acidic residues Keap1 pocket involved in binding of Neh domain of Nrf2. Additionally, soy-derived bioactive peptides were applied to investigate the possible molecular mechanism of resistance training and D-galactose injection treatments on the aging model of rat skeletal muscle. Soy-derived bioactive peptides could initiate the change of the mRNA expression of α-actin, IIa, IIx, MHC-I and MHC-IIb, which could effectively inhibit the aging process of rat skeletal muscle.

Influence of fruits-derived bioactive peptides on inflammation in skeletal muscle

Fruits, as the best sources of fibers, vitamins (mainly vitamins C and vitamins K) and minerals (especially potassium), have momentous positive impacts on enhancement of mental and physical health, which can prevent various non-communicable diseases including cardiovascular disease, neurological disease, osteoarthritis, obesity, diabetes mellitus and some types of cancers (Septembre-Malaterre, Remize, and Poucheret 2018; Dong and Xiao 2017). With the advancement of science and technology, fruits-derived bioactive peptides, as potential therapies, have attracted more and more attention in the field of metabolic syndromes and bodily disorders, which also have been isolated from coconut, pineapple, mango, banana and others. Coconuts, as the most key economic crops in the subtropics and tropics, have attracted more deeply exploration due to the nutritional potential. To better understand the functional and biological characteristics of single protein or proteome of coconut, Ma et al. (2022) identified and characterized coconut meat proteins through shotgun proteomic methods as well as combined with various protein-based bioinformatics analysis to reveal the potential functions of bioactive peptides of globulin and antioxidant protein from coconut. For the globulins and antioxidant protein after digesting by pepsin, a total of 11 and 17 potential bioactive peptides were selected, respectively, which were applied as emulsifiers and antioxidants. Pineapple can be used as the source of peptides, proteins and other bioactive compounds, including several valuable proteolytic enzymes such as bromelain. Mala, Sadiq, and Anal (2021) extracted the pineapple-derived bioactive peptides by microwave-assisted extraction and ultrasonic-assisted extraction. According to the determination of molecular weight, the main pineapple-derived bioactive peptides in the extracts were located in the ~23kDa, corresponding to bromelain. Mango is considered to be the best source of protein in tropical fruits and involves in defense, which contains between 0.36 and 0.40%. Karthiraj, Harish Babu, and Senthil Kumar (2020) effectively separated the α-amylase inhibitory peptide from the mango seed hydrolysate for deep eutectic solvents formed with maltose and carnitine. Banana is the prominent food crop in worldwide, together with wheat,

rice and corn, which can provide multiple benefits such as effective antioxidant, hypocholesterolemic, antihypertensive and anti-diabetic. A total of 10 banana-derived bioactive peptides were discovered by Ferreras et al. through ammonium sulfate precipitation, dialysis as well as gel filtration chromatography to extract and purify (Ferreras et al. 2021). There were 8 banana-derived bioactive peptides as angiotensin converting enzyme inhibitor peptides and 2 banana-derived bioactive peptides as the angiotensin converting enzyme antioxidant and inhibitor peptides. This study successfully explained the health beneficial effects associated with banana consumption.

Recommended nutritional management and its combination with drug therapy during functional recovery phase

Functional recovery phase could be defined as the structural and intentional activation of injured muscles as well as also known as the recovery of participation. At recovery process, a nutrient rich diet was required to maintain a strong immune system and to tolerate the demands of high-intensity training, which should be sufficient to meet the potential energy, carbohydrate, proteins and micronutrient requirements to maintain health and performance (Sato 2018). Nutrition intake was necessary for the wound healing and promoting tissue repair in response to muscle injury. Immobilization caused the muscle loss, which was due to the decrease of basal muscle protein synthesis and the increase of negative muscle protein balance period caused by resistance to anabolic stimuli (such as protein intake) (Ahmed, Sun, and Udenigwe 2022). Resistance exercise during functional recovery phase would increase the synthesis of muscle protein and restore the sensitivity to the stimulation of anabolism. Isenmann et al. (2019) investigated the effects of carbohydrates and proteins on the regeneration and physical performance after exercise. After high-intensity training, a total of 35 male participants ingested nothing (control), a glucose shake/protein (shake) or a combination of sour milk cheese and white bread (food) according to a randomized crossover design. Administration of protein/carbohydrate after exercise could reduce the excretion of pro-inflammatory cytokines, but rise the excretion of anti-inflammatory proteins in serum, which might be explained as anti-inflammatory effect. In fact, skeletal muscle recovery was related to the increased expression of IL-10 and the decreased expression of TNF-alpha in animal model where muscle injured was induced by the administration of notoxin. The complete inhibition of inflammation after high-intensity training (such as after glucocorticoid administration) promoted the catabolic effect of skeletal muscle, which could be interpreted as an indication of pro-regenerative influences (Sato 2022).

Additionally, the efficiency of muscle anabolism varied with protein consumption due to different types of protein sources. Compared with common animal derived protein sources casein and whey, Koshinaka et al. (2021) examined the effect of egg white protein feeding on skeletal muscle injury in rats. Egg white protein, as an arginine-rich protein source, could accelerate skeletal muscle gain. The higher arginine amount in egg white protein was one of the reasonable explanations for its stronger muscle enhancing effects on skeletal muscle, as arginine in egg white protein stimulates the secretion of anabolic hormones. Even though, the possible mechanism of stronger muscle gain effects induced by egg white protein could not be detected in the mTOR and insulin signals. Compared with other common animal proteins, egg white protein might have higher efficiency in muscle enhancement (Wang et al. 2022b). Nevertheless, the serious muscle injuries were mainly due to the subsequent inadequate adaptation and mechanical stress of the musculoskeletal system, the role of nutrition in the occurrence of serious muscle injuries might seem negligible. It was necessary to recommend nutrition management and its combination with drug treatment during functional recovery. Secreted protein acidic and rich in cysteine was used to study its relationship with the mechanobiology of tendons, tissue maturation, homeostasis, and enthesis development (Wang et al. 2021). Loss of secreted protein acidic and rich in cysteine resulted in an altered mechanoresponse and reduced collagen production under a mechanical load controlled *via* the AKT/S6K pathway, ultimately resulting in compromised load-bearing tendons prone to spontaneous tendon rupture. Clinical relevance was underscored by identification of a heterozygous missense mutation in secreted protein acidic and rich in cysteine, which was associated with tendon and ligament injuries in humans. In addition, Katri et al. (2019) investigated the antinociceptive influences and the protective influences on bone and cartilage of the key bioscience peptides used alone or in combination with nonsteroidal anti-inflammatory drugs naproxen in collagen-induced arthritis rat model of rheumatoid arthritis. The key bioscience peptides with nonsteroidal anti-inflammatory drugs naproxen could (*i*) reduce pain, (*ii*) improve overall health, (*iii*) reduce inflammation, (*iv*) protects the cartilage and bone by the combination of structural effects and anti-inflammatory. The study provided evidence for the combination with drug therapy and improving clinical outcomes with reduced side effects. In the past two decades, the number of approvals for peptide-based therapeutics had increased significantly, which was expected to accelerate in the near future. Large-scale and robust randomized clinical trials, with longer follow-up periods, are required to confirm the effects of different nutritional interventions during functional recovery phase (Giraldo-Vallejo et al. 2023). With the progress of technology, this field was expected to open up a new world in the development of safe and more effective drugs to treat a variety of diseases, including skeletal muscle injuries.

Conclusions and future perspective

In recent years, more and more evidences proved that bioactive peptides play an important role in sports nutrition and muscle injury recovery. Food-derived bioactive peptides could regulate the protein synthesis and degradation through PI3K/Akt/mTOR pathway, affect oxidative stress and cellular

metabolism, reduce inflammation in skeletal muscle by regulating NF-κB pathway, relieve cartilage and functional joint pain in skeletal muscle and improve the tendon properties of connective tissue in skeletal muscle. Arginine-rich peptide and its combination with drug therapy also could promote muscle healing. Hence, food-derived bioactive peptides intake was a good and safe way to improve the skeletal muscle function, which could be applied as potential therapeutic candidates for muscle related diseases.

Nevertheless, with the discovery and in-depth study of multifarious food-derived bioactive peptides, there are still four key problems that restrict the research and application of food derived bioactive peptides. (*i*) Food-derived bioactive peptides with the nontoxic effects as well as high stability penetrate into tissues due to the low molecular weight, which also have a great deal of modulatory and regulatory physiological activities. New food-derived bioactive peptides are still waiting to be discovered and explored. (*ii*) At present, most of the research on food-derived bioactive peptides is still in the laboratory stage. The development and exploration of food-derived bioactive peptides separation methods suitable for industrial production still have great limitations. (*iii*) Food-derived bioactive peptides, as functional foods and medicines, have broad application prospects and may cause allergic reactions in a small number of people. *In vivo* activity and safety tests of peptides must be carried out to confirm that the addition of such peptides to food will not have negative effects, and that food-derived bioactive peptides will remain active when used as food ingredients before large-scale food development. (*iv*) Future studies need to focus on the effects of bioactive peptides on metabolic as well as biomechanical responses in humans, the underlying physiological mechanisms of various food-derived bioactive peptides and the potential effects in different skeletal muscle injuries populations in order to evaluate potential field of applications. It is beneficial for market development and utilization to consider chemical modification of the obtained food-derived bioactive peptides structure to improve its activity and make its performance more stable. Overall, exploration of underutilized protein sources to the application of bioactive peptides to foods could bring multiple benefits to consumers, food processors, nutraceutical industries, and pharmaceuticals industry. With this in mind, the majority of strategies presented here should be adopted with pragmatism.

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Authors' contributions

Chenna Di: investigation, writing-original draft preparation, review and editing, and visualization. Wei Jia: writing-reviewing and editing, project administration, visualization and funding acquisition.

Disclosure statement

All authors declare that they have no conflict of interest.

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